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Optimal Congestion Management in Power Systems

by

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Abstract

This dissertation explores optimal congestion management strategies in DC power systems, focusing on improving grid stability and efficiency amid increasing renewable energy integration. A mixed-integer linear programming (MILP) model is developed to optimize congestion management through load shedding, line switching, and phase angle regulation. Using the modified IEEE RTS 24-Bus System, the study simulates scenarios with high renewable energy penetration, highlighting challenges to conventional power system operations. The results show that strategic congestion management can significantly lower operational costs and maintain system reliability while supporting higher renewable energy generation. The findings indicate that combining multiple techniques effectively mitigates congestion and enhances grid stability. The study concludes with recommendations for future research, emphasizing the development of advanced predictive models and the exploration of new strategies for real-time congestion management. This work lays a foundation for further investigation into sustainable and efficient power system operations in the context of global shifts towards renewable energy.

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As I bring this project to a close, I see it as a testament to collective effort, a finely tuned piece in the broader framework of Operational Research. I hope it lays a foundation for future exploration and innovation in this dynamic field.

Own Work Declaration

I hereby declare that the work presented in this dissertation is entirely my own, and all sources and references have been appropriately acknowledged. No part of this work has been submitted for any other degree or qualification at any institution.

Rasika Vivek Dalvi

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1 Introduction

As economically developing countries around the world are trying to reduce greenhouse gas emissions and combat climate change, significant advancements in power systems are essential. The United States, the United Kingdom, and European Union countries have set ambitious targets to achieve net-zero emissions by 2050, underpinned by detailed strategies and legislation like the British Energy Security Strategy and the Fit for 55 package in the EU [23][27][4]. These type of the strategies are put in place to steer away from fossil fuels and shift towards sustainable, and domestic powered energy systems, harnessed from various renewable sources.

This thesis entails the exploration and optimization of power systems on their way to higher penetrations of renewable energy. Emphasis is put on grid stability and congestion management related challenges in DC power systems. This work contributes to the larger goals of sustainability and energy security, which are important to achieving these ambitious net-zero targets.

1.1 Background and Challenges

The nations of the world have committed to aspirational targets in view to hugely reduce carbon footprint and transition to renewable energy systems. The U.K., for example, has committed plans to up its offshore wind capacity to 40 GW by 2030 as a part of a wider strategy to achieve net-zero carbon emissions by 2050 [27]. Massive expansion such as this represents one of the cornerstones of the U.K. government's plan for moving away from conventional power sources, and bolster its energy security by relying more on domestic, renewable sources of power.

Similarly, the United States has outlined its own aggressive climate goals. The U.S. is aiming to reduce greenhouse gas emissions by 50-52% from 2005 levels by 2030 [4]. This is part of a long-term scheme to achieve a net-zero emissions economy by no later than 2050. To meet these targets, the U.S. is focusing on decarbonizing its electricity sector. This is being done by increasing the deployment of renewable energy technologies, and enhancing energy efficiency across all sectors of the economy.

The European Union's Fit for 55 strategy defines a net reduction of greenhouse gases emissions by at least 55% by 2030, compared to 1990, as a milestone toward climate neutrality by 2050 [23]. Along with these ambitious goals, a range of policies and investments are under implementation in European countries to accelerate the transition toward renewable energy. Most of those currently point toward wind power, solar, and bio-energy.

The kind of targets talked about here demand drastic improvements in grid management and energy storage solutions. Given that countries are rapidly increasing their dependence on intermittent renewable sources, such as wind and solar, there is a need to set up systems that manage these variations of generation effectively. This, in essence, sets a path away from traditional fossil fuels and toward a sustainable resources.

The seamless integration of renewable energy sources into existing power systems presents multiple challenges. First and foremost is the issue of congestion in power transmission systems [32]. As the share of renewable energy increases, the existing transmission infrastructure often becomes insufficient to handle the altered power flow patterns, leading to bottlenecks or congestion [16]. This congestion can cause inefficiencies in the system, such as increased operational costs, reduced system reliability, and limits on the maximum exploitable renewable energy capacity.

Another significant challenge is maintaining network security. The power grid must be protected against physical and cyber threats, which can disrupt the supply of electricity and have cascading effects on critical infrastructure. The variability and unpredictability of renewable energy sources add complexity to these security concerns, requiring advanced predictive and real-time monitoring technologies to anticipate and react to potential disruptions.

Additionally, there are technical challenges related to voltage control and frequency stability. Renewable energy sources, particularly wind, do not inherently provide the same level of grid management - such as voltage support and frequency regulation - as conventional thermal power plants. This can lead to issues with power quality and reliability unless appropriate compensatory measures are implemented.

Economic challenges also arise from the need for significant investment in both new generation capacities and grid infrastructure to accommodate renewable energy sources. Furthermore, the traditional business models of utilities are challenged by the decentralization of energy production, necessitating new approaches to pricing, market design, and regulation.

1.2 Problem Statement

The incorporation of renewable energy sources into direct current power systems presents several challenges regarding their variable and to some extent unpredictable nature. Due to this, an inherent risk for power supply fluctuation is bred, which might cause grid congestion and instability [10]. As the penetration of renewables like solar and wind energy is on the rise, power systems are required to transit from the traditional stable configuration to one that is dynamic and adaptive.

Conventional grids were built around predictable centralized power generation and therefore, those are not adequate for handling distributed and unsteady power outputs from renewable resources. This characteristic of renewable sources requires smart approaches for grid management. This shall be done to ensure peak and varying demands are met while safeguarding system stability [29].

To address these problems, congestion management techniques are essential. These techniques include load shedding, line switching, and implementation of phase angle regulators (PAR). These strategies act as excellent tactic for maintaining a balance between supply and demand of power. Furthermore, the application of Mixed-Integer Linear Programming (MILP) model provides a robust framework for optimizing the congested network, while also ensuring effective renewable energy integration.

1.3 Objectives of the study

Outlined below are the objectives that will direct the project's activities and contribute to achieving the desired outcomes.

- **Advance Congestion Management Techniques:** Develop and implement a mathematical model that effectively mitigates congestion in DC power systems. This model seeks to incorporate combination of various congestion management actions such as load shedding, line switching, and phase angle regulator (PAR).
- **Minimize Operating Costs:** Minimize the costs associated with power generation and the additional costs linked to implementation of congestion management techniques.
- **Maximize Renewable Integration:** Develop strategies to increase the integration of renewable energy sources into the grid, aligning with national sustainability goals.
- **Enhance Grid Stability:** Use the MILP model to manage power flow variations from renewable sources, ensuring consistent and reliable power delivery.

1.4 Contributions

This study introduces an innovative approach by combining multiple congestion management strategies into a single Mixed-Integer Linear Programming (MILP) model. These strategies include load shedding, line switching, and Phase Angle Regulators (PAR). By integrating these methods, the study addresses grid stability and efficiency in scenarios with a high level of renewable energy integration.

This new model offers a comprehensive solution that has not been fully explored in existing research.

The research further applies this innovative MILP model to a specifically adapted IEEE RTS 24-Bus System. This system has been modified to reflect high renewable energy integration, making the simulations close to real-world applications. This practical implementation assists in testing and analyzing how effective these congestion management strategies are under such conditions.

Another major focus of the research is a detailed analysis of the cost-effectiveness and reliability of different congestion management strategies. Through thorough simulations, the study identifies which strategies are most economical while enhancing the reliability of power systems. This analysis aids in understanding which methods achieve the best balance between cost and performance.

The dissertation also develops a new framework that enhances decision-making in managing congestion. This framework offers key insights that help power systems efficiently incorporate more renewable energy without affecting grid stability.

These contributions clearly demonstrate the progress this research brings to managing power systems, highlighting valuable applications, innovative methods, and significant implications for future studies and practices in this field.

1.5 Structure of the Dissertation

The structure of the following sections of the dissertation is laid forth in a manner to give a comprehensive overview of power system congestion management. Following the introduction, the dissertation is divided into several key sections:

- **Literature Review:** This section will review existing research on renewable energy integration and grid management strategies. It will provide background information on the existing approaches to the problem at hand.
- **Problem Formulation and Solution Methods:** This section will be devoted to explaining, in some reasonable detail, the mathematical model developed to deal with grid congestion and grid security-related issues. It will include all sets, variables, and equations used in the MILP model.
- **Model Implementation:** This section will describe how the model is implemented on the modified IEEE RTS 24-Bus System. It discusses how the formulation of the Congestion Management Model integrates Load Shedding, Line Switching, and PAR. The details of the software and tools used for implementation will also be given.
- **Results and Analysis:** This section will depict the results obtained after the implementation of this model and analyze the effectiveness of the proposed solutions on different combinations of congestion management strategies.
- **Conclusion and Future Works:** The final section is reserved for summarizing the general outcome of the study and recommending areas that would present better avenues for future research. The contributions made toward the integration of renewable energy into the grid and its management will be highlighted.

The sections have been designed so that the final one closes them by providing a set of practical recommendations that can be implemented to improve power grid operation with the increasing renewable energy integration. This structured approach ensures a logical flow of information and facilitates a clear understanding of the research contributions.

2 Literature Review

In recent years, DC power systems have attracted more attention because of their effectiveness and, most importantly, the role they play in integrating renewable sources of energy. Modern power systems, in their evolution to serve an ever-increasing consumer demand, highlight the important role that DC grids play in congestion management. Unlike AC systems, some of the advantages of DC systems are lesser power loss over long distances and better interfaces with renewable sources of energy. These attributes make the DC grids quite important for modern power systems and also help manage congestion effectively [12].

Congestion management in DC power systems is a matter of grid reliability and security. Congestion reasons are complex, relating to physical and operational issues, as well as market problems. It is necessary that the challenges be addressed with full knowledge of techniques at hand toward the management of congestion. This section gives an overview of these techniques and in-depth analysis of load shedding, line switching, and phase angle regulators related to operational research in power systems.

2.1 Congestion Management Techniques Overview

Congestion management in power systems is a tedious task which requires the application of several strategies oriented to address specific operational contexts and grid conditions. Among the various techniques available, we will focus on those most relevant to our model, specifically Load Shedding, Line Switching, and the use of PAR. These techniques have been tried and tested in fighting congestion, maintaining grid stability, and optimizing power flow within a network [22]. These approaches can significantly improve the resilience and efficiency of power systems with increasing demand and integration with renewable energy sources. Following the discussion of these key methods, we will briefly outline other widely used congestion management techniques that, while not directly implemented in our model, play an essential role in broader power system operations [33].

Load Shedding

Load shedding is a versatile tool for managing congestion in power systems, with various methods tailored to different challenges. Load shedding is often seen as a last resort, but it is an effective way to maintain system stability and avoid widespread blackouts.

One approach uses the Grey Wolf Optimizer, a metaheuristic algorithm that mimics the social structure of wolves to find the best load shedding strategy [25]. This method is effective in dealing with complex and non-linear congestion problems, ensuring system stability while minimizing disruptions. Another method takes a decentralized approach, allowing local substations to make quick decisions on load shedding [22]. This reduces the reliance on central control, enabling faster responses and minimizing the impact on consumers.

Other strategies include priority-based load shedding, where loads are ranked and selectively shed based on their importance, helping maintain grid balance [33]. In DC microgrids, a hierarchical control structure sheds non-critical loads first, adapting to real-time changes to keep the microgrid stable [17][15]. Another method uses the power network's topology, employing real-time data to predict and prevent potential congestion points through preemptive load shedding [34]. Lastly, linear programming techniques are used in some approaches to optimize load shedding, ensuring it is done efficiently with minimal disruption to consumers [10]. These varied methods showcase how load shedding can be tailored to effectively manage congestion across different power systems.

However, load shedding is a delicate matter that should be managed well to avoid unnecessary disruptions. Excessive reliance on load shedding may result in economic losses and consumer confidence. Integrating advanced metering and control systems can improve the precision and responsiveness of

load shedding programs, making them more effective in mitigating congestion without significant drawbacks.

Line Switching

Line switching is a dynamic congestion management technique that seeks to alter the physical configuration of the grid and, with that, create a redesign of power flows [24].

Transmission switching has been employed in various innovative ways to manage congestion in power systems. One effective approach involves a heuristic algorithm that prioritizes switching actions based on minimizing load shedding while alleviating congestion [9]. This method significantly enhances system reliability by reducing the need for load shedding, thus minimizing economic losses from forced outages. By focusing on critical lines that impact congestion the most, this strategy achieves substantial relief without requiring extensive computational resources, making it both effective and resource-efficient [9].

Another approach decentralizes the decision-making process, allowing local grid operators to switch lines based on real-time data and localized congestion signals [11]. This method improves the responsiveness of congestion management, particularly in large, interconnected grids where centralized control might be slow or prone to bottlenecks. Additionally, transmission switching has been integrated into multi-objective optimization frameworks and security-constrained models [26]. These frameworks balance cost minimization with grid reliability, ensuring that switching decisions are both economically sound and secure. By considering multiple scenarios and adhering to the N-1 contingency criterion, these approaches prevent new vulnerabilities while resolving congestion, proving particularly valuable in complex and heavily loaded power systems [20][5].

Though the line-switching congestion control method is quite effective, it requires sophisticated control systems that prevent it from turning into grid destabilization. Further, frequent switching may also cause additional wear and tear on equipment, due to which planning and proper coordination with other congestion management techniques, like load shedding and generation re-dispatch, becomes necessary.

Phase Angle Regulator

Phase angle regulators are a quite critical congestion management tool in DC power systems. The angles can be altered to shift power away from the congested area and re-balance the load across the network, retaining its stability [3].

One method involves using a fuzzy logic control algorithm to optimize PAR operations [3]. This approach dynamically adjusts the phase angle based on congestion levels, allowing real-time control to alleviate congestion effectively. The fuzzy logic system's strength lies in its flexibility, considering multiple factors simultaneously and enabling responsive adjustments to maintain grid stability. Another approach incorporates PAR into a multi-objective optimization framework where it coordinates multiple phase shift transformers to minimize costs [14]. Additionally, some studies integrate PSTs into a security-constrained optimization framework, adhering to the N-1 contingency criterion to maintain system security even during failures [30]. PARs are also used to fine-tune phase angles, preventing overloads in critical network components, especially in systems with variable loads [32]. Reviews highlight the importance of using PARs and PSTs to redirect power to underutilized paths, emphasizing their integration with other control systems for optimal congestion management [12].

However, the cost and requirements of sophisticated control systems make PAR application still challenging. They require strategic placement for maximum benefits and operation planning so they would not disrupt the grid dynamics to create new congestion points [30].

Few Other Techniques for Congestion Management

There are a good number of congestion management techniques at one's disposal when it comes to power systems, each with its own merits and areas of application. In our model, we will implement only a few that are relevant for our objectives and operational context. Nevertheless, it is worth mentioning other techniques that are equally effective and extensively used, where grid stability may be affected by them.

For instance, Generation Re-dispatch involves realigning generators' output in order to shift power flows in the network, and therefore freeing congested areas [21]. It is, therefore a very commonly applied technique employed in both DC and AC grids towards ensuring system balance and prevention of overloads. Generation re-dispatch is particularly valuable in scenarios where quick adjustments are necessary to avoid system instability [17][2].

The second technique involves Dynamic Thermal Rating, which is based on the running of thermal limits for transmission lines dynamically, based on actual environmental conditions instead of relying on static ratings[31]. Therefore, it is going to allow efficient utilization of the presently existing infrastructure that is available, thus giving maximum capacity to the transmission lines whenever the conditions are at will, such as lower ambient temperatures or higher wind speeds. DTR is particularly useful in grids where congestion is frequent since more capacity is made available with minimal infrastructure upgrades [31].

Bus Splitting, lastly, is an operation whereby a bus is split into several independent substations. The objective of this operation is to handle the distribution of load better and, in effect, handle congestion better. This technique is very important when used in complex networks which reduces substantial risks of congestion due to slight changes in the paths of the flow [31][6].

2.2 Mathematical Modelling in Congestion Management

Mathematical models are going to be of essence in the study and application of the techniques for Congestion Management. These models provide a framework for analyzing complex power systems, predicting possible congestion points, and the optimization of the application of the various management strategies. These models mainly include linear and mixed integer programming which are focus of this study [9]. Few advanced models involve heuristic and meta-heuristic algorithms [13]. Time-series models are implemented for multi-period scenarios where the demand and supply changes over time [26].

While mathematical models lay the theoretical groundwork, methodologies are the practical applications of these models, encompassing specific techniques and algorithms used to manage congestion effectively.

- **Optimal Power Flow (OPF) Models:** OPF models are essential for determining the most efficient power flow through the system under various constraints [7][1]. They help in reducing generation costs subject to the system operation within technical limits.
- **Security-Constrained OPF (SCOPF) Models:** SCOPF extends OPF by incorporating security constraints, ensuring that the system can withstand component failures without leading to cascading outages. This includes N-1 contingency planning [20]. This model thus becomes very important in system reliability, especially for highly renewable energy-penetrated grids.
- **Stochastic Models:** Stochastic models account for the uncertainties in power system operations, such as variable renewable energy output. Stochastic models help operators deal with such risks in unpredictable generation and load patterns and consequently derive decision-making even in real-time operations [13].

- **Decentralized and Distributed Models:** To manage congestion in large-scale or geographically dispersed power systems, decentralized models that allow for localized decision-making have been developed. These models facilitate faster response times and reduce the computational burden on a central controller, enhancing the scalability and efficiency of congestion management solutions [11].

Mathematical models in congestion management serve as critical tools for ensuring the efficient and reliable operation of power systems. By leveraging various mathematical and computational techniques, these models help system operators anticipate, analyze, and mitigate congestion. Hence, maintaining grid stability and preventing service interruptions. The continuous development and refinement of these models are essential as power systems evolve towards more sustainable and dynamic configurations, with increased penetrations of distributed and renewable energy resources.

2.3 Research Gaps

Despite significant progress in congestion management research, several gaps remain that need to be addressed to enhance the effectiveness of these techniques in modern power systems:

1. **Combination of the Various Techniques:** All of the congestion management techniques have been very well studied in individual applications. However, not a lot of research work has been done on the integration of these techniques together to come out with a more coherent and integrated management strategy [26]. Future research should focus on the development of models combining techniques like load shedding, line switching, and PAR to achieve optimum management of congestion.
2. **Real-Time Application and Scalability:** Many existing studies focus on theoretical or simulated environments. Therefore, real-time application of the techniques of congestion management is an area for further study, especially for large, interlinked power systems. Knowing how the techniques will perform under real-world conditions makes them very important if used correctly.
3. **Economic Impacts and Market Structures:** The economic implications of congestion management strategies, particularly in deregulated markets, are not fully understood. More work should be done on how these strategies would affect prices in markets, infrastructure investment, and the financial viability of renewable projects.
4. **Technological Advancements:** Emerging technologies such as artificial intelligence and machine learning hold promise for improving congestion management [13][25]. However, their application is still in the early stages, with much yet to be proved.
5. **Impact of Renewable Energy Variability:** Much research has gone into the integration of renewables, but focused studies are still required on managing variability and intermittency [17] [29]. More importantly, advanced predictive models and newer strategies to mitigate these impacts have to be developed in order to maintain grid stability.

Addressing these gaps is critical for advancing the field of congestion management and ensuring the reliability and efficiency of future power systems.

3 Problem Formulation and Solution Methods

The proposed dissertation tackles the significant challenges arising from the rapid adoption of renewable energy sources and the slower pace of upgrades in transmission network capacities. It seeks to develop a strategic framework for managing congestion within DC power systems effectively. This section presents the conceptual basis of the mathematical model devised to address the complexities of modern grid management, with a strong emphasis on both cost efficiency and system reliability.

At the core of this dissertation lies the development of a comprehensive mathematical model aimed at optimizing congestion management strategies while ensuring the smooth integration of renewable energy sources into the DC power system. As the power grid increasingly relies on renewable energy, traditional congestion management methods fall short of addressing the complexities introduced by these new energy sources. To bridge this gap, the model incorporates advanced techniques such as load shedding, line switching, and the application of PAR. Each of these techniques brings different impacts, costs, and implications for grid reliability and security, necessitating a balanced approach.

The key objective of this model is to identify the most cost-effective strategies for managing congestion within the grid. This involves minimizing the total operational costs linked to power generation, including the expenses associated with the deployment of various congestion management measures. By achieving this cost reduction, the model not only seeks to improve the economic efficiency of the power grid but also to enhance its stability and reliability, particularly in response to the varying demand patterns characteristic of modern energy consumption. Thus, the dissertation provides a strategic approach that addresses both the economic and operational challenges posed by the increasing integration of renewable energy into the power grid.

3.1 Mathematical Model Formulation

This subsection will provide a detailed explanation of the development of the mathematical model, focusing on how the congestion management problem is represented as a mixed-integer linear programming (MILP) model [18]. The MILP model is structured to incorporate a variety of decision variables, which include the output levels of generators, the operational status of transmission lines (considering on/off states for line switching), the amounts of load shedding, and the phase shift angle. These variables play a crucial role in the model's ability to capture the complex dynamics of congestion management within the power grid [18].

The formulation of the objective function within this model is particularly critical, as it is designed to minimize the total operational cost. This cost includes all expenses associated with the implementation of different congestion management techniques and actions, such as load shedding, line switching, and adjusting the phase shift angle. By effectively minimizing these costs, the model aims to optimize the economic efficiency of the power grid while maintaining system reliability.

Through this comprehensive MILP approach, the model provides a robust framework for addressing the challenges of congestion management in modern power systems, particularly in the context of increasing reliance on renewable energy sources. The detailed development of the model in this subsection will lay the groundwork for its application in real-world scenarios, ensuring that it is both theoretically sound and practically viable for enhancing the overall performance of the power grid.

Sets

G : Set of generators.

B : Set of buses.

L : Set of lines.

L_o : Set of ordinary lines without line switching and PAR ability. $L_o \subseteq L$.

L_p : Set of lines that can have PAR. $L_p \subseteq L$.

L_s : Set of lines that can be switched. $L_s \subseteq L$.
 $L = L_o \cup L_p \cup L_s$ and $L_o \cap L_p \cap L_s = \emptyset$

Parameters

P_b^D : Power load at bus b (MW).
 P_g^{G+} : Maximum power generation capacity of generator g (MW).
 P_g^{G-} : Minimum power generation capacity of generator g (MW).
 C_g^G : Cost of running generator (£/MWh)
 P_l^{L+} : Maximum power flow capacity of line l (MW).
 X_l : Reactance of line l (MW^{-1}).
 C_b : Reward for per unit of load supplied at each bus b (£)
 C_l^L : Cost of line switching per line l (£)
 C_p : Cost of installing PAR per line l (£)
 M : A sufficiently large number for the line switching constraints.
 $\alpha_l^{\min}$: Lower limit of phase shift introduced by the PAR on line l
 $\alpha_l^{\max}$: Upper limit of phase shift introduced by the PAR on line l
 a_{bl} : Element of Node/Arc incidence matrix A

$$a_{bl} = \begin{cases} -1 & \text{if } b \text{ is the start bus of line } l \\ +1 & \text{if } b \text{ is the end bus of line } l \\ 0 & \text{otherwise} \end{cases}$$

Variables

p_g^G : Power output of generator g (MW).
 p_l^L : Power flow in line l (MW).
 δ_b : Voltage angle at bus b
 k_b : Variable that controls the proportion of the load P_b^D supplied at each bus b .
 α_l : Phase shift introduced by the PAR on line
 y_l : Binary variable for line l to switch on or off (1 if line l is on, 0 if off)
 w_1 : Binary variable indicating if load shedding is considered (1 if active, 0 otherwise)
 w_2 : Binary variable indicating if line switching is considered (1 if active, 0 otherwise)
 w_3 : Binary variable indicating if PAR is considered (1 if active, 0 otherwise)

Objective function

The objective function is designed to minimize the total operational cost of the power system, considering various components of costs associated with generation, load shedding, line switching, and the use of PAR. Each term in the function corresponds to a specific operational or remedial action. The decision variables, w_1 , w_2 , and w_3 indicate whether the remedial actions such as load shedding, line switching, and PAR are considered, respectively.

Generation Cost: This is the first term in equation (3.1). This term represents the cost of generating electricity at each generator. The cost is a product of the power output by each generator and its specific cost of generation.

Load Shedding Cost: This is the term associated with w_1 in equation (3.1). This term accounts for the economic penalties associated with not meeting the full electricity demand at each bus. It multiplies the unserved power at each bus by a penalty factor, reflecting the severity of load shedding. Load shedding cost is applied when w_1 is active.

Line Switching Cost: This is the term associated with w_2 in equation (3.1). This term includes the costs associated with turning transmission lines on or off. Line switching cost is applied when w_2 is

active.

Phase Angle Regulator Cost: This is the term associated with w_3 in equation (3.1). This cost is associated with the use of PAR, which adjust the phase angles across transmission lines to manage power flows and alleviate congestion. PAR cost is applied when w_3 is active.

$$\min \sum_{g \in G} C_g^G p_g^G + w_1 \sum_{b \in B} C_b(1 - k_b)P_b^D + w_2 \sum_{l \in L_s} C_l^L(1 - y_l) + w_3 \sum_{l \in L_p} C_p |\alpha_l| \quad (3.1)$$

The constraint (3.2) makes sure that the phase shift α_l introduced by the PAR on line l is confined within the bounds set by $\tilde{\alpha}_l$.

$$-\tilde{\alpha}_l \leq \alpha_l \leq \tilde{\alpha}_l, \forall l \in L_p \quad (3.2)$$

Hence the objective function can be written as follows:

$$\min \sum_{g \in G} C_g^G p_g^G + w_1 \sum_{b \in B} C_b(1 - k_b)P_b^D + w_2 \sum_{l \in L_s} C_l^L(1 - y_l) + w_3 \sum_{l \in L_p} C_p \tilde{\alpha}_l \quad (3.3)$$

Constraints

Following constraints will include power flow equations, capacity limits, linear relaxation constraints and other operational constraints necessary to ensure the physical and safe operation of the power system.

Action Selection Constraint:

The action combination selection constraint ensures that the combination of different congestion management techniques does not exceed the operational bounds.

$$0 \leq w_1 + w_2 + w_3 \leq 3 \quad (3.4)$$

The constraint (3.5) specifies that the variables w_1 , w_2 , and w_3 which indicate whether the congestion management action such as load shedding, line switching and phase angle regulation are active (i.e. $w = 1$) or not (i.e. $w = 0$) respectively. The variables w_1 , w_2 , and w_3 can only take binary values, ensuring clear operational statuses for all congestion management actions considered in the model.

$$w_1, w_2, w_3 \in \{0, 1\} \quad (3.5)$$

Kirchhoff Current Law (KCL):

This constraint ensures that at each bus, the sum of incoming and outgoing power, including generation and power flows on transmission lines, equals the effective demand at that bus after accounting for any load shedding [18].

$$\sum_{g \in G} p_g^G + \sum_{l \in L} a_{bl} p_l^L = k_b P_b^D, \forall b \in B \quad (3.6)$$

Kirchhoff Voltage Law (KVL):

Kirchhoff's Voltage Law (KVL) is applied to ensure that the sum of electrical potential differences around any closed network loop is zero [18]. This law is crucial for maintaining voltage stability in a power distribution system. The KVL constraints are formulated differently depending on the type of line, such as ordinary lines, lines with switching, and lines with PAR.

KVL for ordinary lines: For lines without any additional operational complexities like switching or PAR, the KVL is straightforward. The constraint (3.7) shows the KVL constraint for such ordinary

lines.

$$X_l p_l^L + \sum_{b \in B} (a_{bl} \delta_b) = 0 \quad , \forall l \in L_o \quad (3.7)$$

KVL for line switching: For lines where switching is considered, the KVL must account for the possibility that a line being disconnected. When a line is switched off (i.e., $y_l = 0$), the power flow through that line is effectively zero, and so should be the voltage across it. The introduction of a large number M ensures that when the line is switched off, the equation is trivially satisfied, and no voltage drop occurs across the line.

$$-(1 - y_l)M \leq X_l p_l^L + \sum_{b \in B} a_{bl} \delta_b \leq (1 - y_l)M \quad , \forall l \in L_s \quad (3.8)$$

KVL for PAR: For lines equipped with phase angle regulator, KVL needs to consider the additional phase shift α_l introduced by the PAR. This phase shift allows for greater control over the flow of electricity and the voltage levels across the network by adjusting the phase angles directly.

$$X_l p_l^L + \sum_{b \in B} (a_{bl} \delta_b + \alpha_l) = 0 \quad , \forall l \in L_p \quad (3.9)$$

Generator Limits:

These constraints ensure that the power output of each generator remains within its specified minimum and maximum limits [2].

$$P_g^{G-} \leq p_g^G \leq P_g^{G+} \quad , \forall g \in G \quad (3.10)$$

Line Flow Limits:

The constraints (3.11) and (3.12) ensure that the power flow in each line does not exceed its maximum capacity in either direction. The constraint varies depending on whether a line is operating normally or it has switching functionality.

$$-P_l^{L+} \leq p_l^L \leq P_l^{L+} \quad , \forall l \in L_o, \forall l \in L_p \quad (3.11)$$

$$-y_l \cdot P_l^{L+} \leq p_l^L \leq y_l P_l^{L+} \quad , \forall l \in L_s \quad (3.12)$$

The constraint (3.13) specifies that the variable y_l , which indicates whether line l is switched on ($y_l = 1$) or off ($y_l = 0$), can only take binary values, ensuring clear operational statuses for all lines that have switching functionality.

$$y_l \in \{0, 1\} \quad , \forall l \in L_s \quad (3.13)$$

Load Proportion Constraint:

Determines the proportion of load shedding that can occur at each bus, constrained by whether load shedding is enabled or not.

$$0 \leq k_b \leq 1 \quad , \forall b \in B \quad (3.14)$$

Linear Relaxation for Load Shedding:

The following constraint (3.15) represents linear relaxation for load shedding proportion and load shedding action.

$$1 - w_1 \leq k_b \quad , \forall b \in B \quad (3.15)$$

Linear Relaxation for Line Switching:

Manages the operational status of each line, indicating whether a line is on or off based on whether line switching action is active or not. The following constraint also represents linear relaxation for

switching operation of line and line switching action.

$$1 - w_2 \leq y_l \quad , \forall l \in L_s \quad (3.16)$$

Phase Shift Constraint:

The constraint (3.17) controls the range of phase shift introduced by the PAR. It is bound by minimum and maximum limits modulated by whether installation of PAR action is considered.

$$w_3 \cdot \alpha_l^{min} \leq \alpha_l \leq w_3 \cdot \alpha_l^{max} \quad , \forall l \in L_p \quad (3.17)$$

Next, the complete mathematical problem formulation will be presented in a unified manner. This section will provide an overview that consolidates the various elements of the model, which have been explained in detail earlier, including the constraints and the objective function. By bringing together these components, the overall structure of the mathematical model can be clearly understood, offering a concise yet comprehensive summary of how the congestion management problem is approached. This consolidated formulation serves as a quick reference, allowing for an efficient review of the entire model before delving into its application and analysis.

$$\begin{aligned} \min \quad & \sum_{g \in G} C_g^G p_g^G + w_1 \sum_{b \in B} C_b(1 - k_b)P_b^D + w_2 \sum_{l \in L_s} C_l^L(1 - y_l) + w_3 \sum_{l \in L_p} C_p \tilde{\alpha}_l \\ \text{subject to} \quad & \end{aligned}$$

$$-\tilde{\alpha}_l \leq \alpha_l \leq \tilde{\alpha}_l \quad , \forall l \in L_p$$

$$\text{Action Selection Constraint: } 0 \leq w_1 + w_2 + w_3 \leq 3$$

$$w_1, w_2, w_3 \in \{0, 1\}$$

$$\text{KCL: } \sum_{g \in G} p_g^G + \sum_{l \in L} a_{bl} p_l^L = k_b P_b^D \quad , \forall b \in B$$

$$\text{KVL for ordinary lines: } X_l p_l^L + \sum_{b \in B} (a_{bl} \delta_b) = 0 \quad , \forall l \in L_o$$

$$\text{KVL for switching lines: } -(1 - y_l)M \leq X_l p_l^L + \sum_{b \in B} a_{bl} \delta_b \leq (1 - y_l)M \quad , \forall l \in L_s$$

$$\text{KVL for PAR lines: } X_l p_l^L + \sum_{b \in B} (a_{bl} \delta_b + \alpha_l) = 0 \quad , \forall l \in L_p$$

$$\text{Generator limits: } P_g^{G-} \leq p_g^G \leq P_g^{G+} \quad , \forall g \in G$$

$$\text{Line flow limits: } -P_l^{L+} \leq p_l^L \leq P_l^{L+} \quad , \forall l \in L_o, \forall l \in L_p$$

$$-y_l \cdot P_l^{L+} \leq p_l^L \leq y_l P_l^{L+} \quad , \forall l \in L_s$$

$$y_l \in \{0, 1\} \quad , \forall l \in L_s$$

$$\text{Load Proportion: } 0 \leq k_b \leq 1 \quad , \forall b \in B$$

$$\text{Load Shedding Relaxation: } 1 - w_1 \leq k_b \quad , \forall b \in B$$

$$\text{Line Switching Relaxation: } 1 - w_2 \leq y_l \quad , \forall l \in L_s$$

$$\text{Phase Shift Constraint: } w_3 \cdot \alpha_l^{min} \leq \alpha_l \leq w_3 \cdot \alpha_l^{max} \quad , \forall l \in L_p$$

3.2 Techniques for solving the mathematical model

The mathematical model developed for congestion management in power systems is designed with flexibility, allowing the user or operator to choose between two distinct approaches for solving the optimization problem. These approaches revolve around how the congestion management action indicators, denoted as w_1 , w_2 , and w_3 , are treated within the model. The indicators represent different congestion management strategies and can be utilized either as decision variables or as input parameters, depending on the specific needs and objectives of the user.

3.2.1 Treating w 's as Decision Variables

In the first approach, the congestion management action indicators w_1 , w_2 , and w_3 are treated as decision variables within the optimization problem. When this approach is selected, the model actively determines the optimal values for these indicators as part of the solution process. This means that the model is responsible for identifying the best combination of congestion management strategies that minimizes the total operational costs while adhering to the system constraints.

Using w 's as decision variables transforms the problem into a mixed-integer programming (MIP) problem. The complexity of this approach arises because the indicators w_1 , w_2 , and w_3 are binary variables that can take on values of 0 or 1. These binary variables are then multiplied by other decision variables in the objective function, leading to a mixed integer quadratic programming problem. The quadratic nature of this problem reflects the interactions between the congestion management strategies and their impact on the overall system.

The primary advantage of this approach is that it allows the model to explore a wide range of potential solutions and identify the optimal combination of congestion management strategies automatically. This is particularly useful in complex systems where manual determination of the best strategies would be time-consuming or impractical. By using w 's as decision variables, the model can dynamically adjust to varying system conditions and provide an optimal solution tailored to the specific problem at hand.

3.2.2 Treating w 's as Input Parameters

The second approach involves treating the congestion management action indicators w_1 , w_2 , and w_3 as input parameters rather than decision variables. In this scenario, the user or operator manually sets the values for these indicators before solving the optimization problem. This approach requires the user to predetermine the combination of congestion management strategies they wish to evaluate.

When w 's are used as input parameters, the mathematical problem simplifies into a linear programming (LP) problem. The linear nature of this problem results from the fact that the indicators w_1 , w_2 , and w_3 are fixed, and their values do not interact multiplicatively with other decision variables in the objective function. Instead, the model focuses on optimizing the remaining decision variables, given the fixed values of w_1 , w_2 , and w_3 .

The advantage of this approach lies in its simplicity and clarity. By manually setting the values for w 's the user can isolate and examine the impact of specific congestion management strategies on the system's performance. This is particularly beneficial for conducting scenario analysis or for situations where the user has prior knowledge or preferences regarding which strategies should be implemented. It allows the user to compare the outcomes of different strategy combinations without the added complexity of a mixed-integer problem. This can be a more labor-intensive process, especially in systems with a large number of possible strategy combinations. Additionally, because the indicators are fixed, the model may not always identify the most cost-effective or optimal combination of strategies, particularly in complex or dynamic systems where interactions between strategies are significant.

3.3 Software Tools and Computational Framework

The computational framework employed in this research was meticulously constructed within a **Python** environment, in conjunction with **IBM ILOG CPLEX Optimization Studio V22.1.1**. This combination of tools provided a robust and scalable platform for solving complex optimization problems related to power system congestion management. The integration of these software tools enabled a comprehensive approach to both modeling and solving large-scale, mixed-integer optimization problems with high levels of efficiency and precision.

Data handling and preprocessing were managed using '**Pandas**' for efficient data manipulation and '**Numpy**' for numerical operations. These libraries allowed for the seamless import, processing, and structuring of complex datasets essential to the model. The core of the optimization model was developed using the '**docplex.mp.model**' module, which is part of the IBM Decision Optimization CPLEX Modeling library. This module was chosen for its ability to facilitate the creation of detailed mathematical models, specifically designed for linear and mixed-integer programming.

The docplex module's flexibility was crucial in handling the mixed-integer programming (MIP) aspects of the model. This allowed the inclusion of binary decision variables alongside continuous ones, which is essential for accurately modeling the on/off states of transmission lines, as well as other binary decisions related to load shedding and phase angle regulation. The inclusion of these mixed-integer elements added a layer of complexity to the optimization problem, which was effectively managed by the CPLEX solver.

The IBM ILOG CPLEX Optimization Studio V22.1.1, accessed through the cplex Python API, was the solver of choice for this research. CPLEX is renowned for its ability to efficiently solve large-scale linear programming (LP), mixed-integer programming (MIP), and quadratic programming (QP) problems. In this research, CPLEX's advanced solving capabilities were particularly valuable for handling the quadratic nature of the objective function, which resulted from the interaction of binary decision variables with other continuous variables. The solver's efficiency ensured that optimal solutions were obtained in a timely manner, even when dealing with the increased complexity of mixed-integer quadratic programming (MIQP).

4 Implementation of the Model

Effective congestion management is crucial for ensuring the stability and efficiency of modern power systems, particularly with the increasing integration of renewable energy sources. In this section, a case study will be presented in detail, focusing on how the mathematical model is implemented with respect to system topology, transmission line data, generator capacities, and relevant economic factors essential for solving the congestion management problem.

4.1 Overview of IEEE RTS 24-Bus System

The IEEE RTS 24-Bus System, also known as the RTS-96 system, serves as an important benchmark for evaluating power system operations and electricity market mechanisms[19]. Originally part of the IEEE Reliability Test System [28], this model has been extensively utilized in both academic and industry research to simulate various aspects of power systems, such as reliability, economic dispatch, and market operations. The version of the IEEE 24-bus system mentioned in [19] incorporates several updates and modifications to better reflect the challenges of modern power systems, including the integration of renewable energy sources like wind power.

This system is a single-area power grid comprising 24 buses, 38 transmission lines, and 10 generator units [28]. These components are interconnected to realistically mimic the complexity of a regional power grid, making it a valuable tool for testing and validating various power system models and strategies [19]. The 24 buses serve as nodes where power is either generated, transmitted, or consumed. Key elements of the system include reference bus, generator buses, and load buses, each playing a vital role in maintaining the grid's overall balance and stability.

The load data utilized in the model implementation is drawn from [28] and specifically reflects the system peak conditions as outlined in Table 10 in IEEE report. This table provides detailed information on bus load data at the time of system peak, with each bus load expressed as a percentage of the total system load. The system's total peak load is 2,850 MW, reflecting realistic operational conditions [28]. The generator units in original system are characterized by a range of technical and economic parameters, including maximum and minimum power outputs, ramp rates, reserve capacities, and operational costs. The transmission network of the 24-bus system consists of 38 lines, each defined by its reactance and capacity [28].

A significant update in [19] system, is the inclusion of six wind farms, strategically located at various nodes throughout the grid. These wind farms, each with a capacity of 200 MW, introduce a layer of complexity by adding intermittent renewable energy sources into the system. In this model, wind farms are located at nodes 3, 5, 7, 16, 21, and 23 [19]. To further enhance the model's realism, the capacity of specific transmission lines connecting nodes (15, 21), (14, 16), and (13, 23) have been decreased to 400 MW, 250 MW, and 250 MW, respectively [19]. This intentional introduction of bottlenecks within the network allows for a more detailed study of congestion and its potential mitigation in a system with significant renewable energy penetration [19]. The combination of these reduced line capacities and the inherent stochastic nature of wind power generation creates a challenging environment for testing and validating various optimization models and strategies.

4.2 Modifications in IEEE RTS 24-Bus System

In this subsection, the key modifications made to the IEEE RTS 24-Bus System for the revised model are outlined. These changes were introduced to reflect a more contemporary power system scenario with a greater emphasis on renewable energy integration and streamlined system design. The system line diagram, shown in fig. 1, visually encapsulates these updates [19]. It illustrates the network topology and highlights the buses where renewable generators are integrated with **W**.

The first significant modification involves the configuration of generator units at the buses. In the

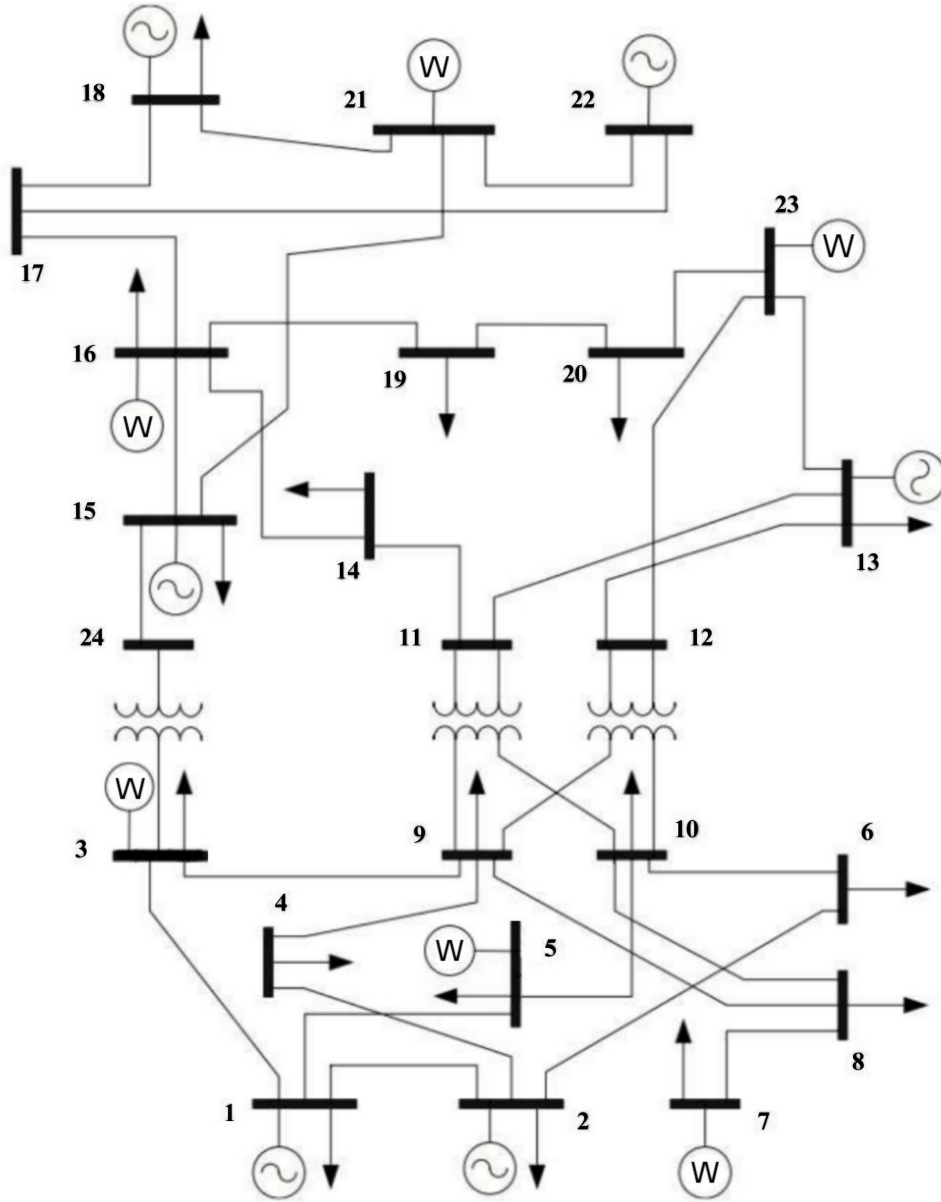


Figure 1: Modified version of IEEE RTS 24-Bus System - Single area

original system, [28], several buses had multiple generators with varying capacities. However, in the revised model, each bus with generation capability is simplified to have only one generator per bus. The capacities of the multiple generators originally present at these buses are aggregated into a single generator unit. This approach not only simplifies the modeling process but also provides a clearer analysis of generation dynamics at each bus.

Another major change in the revised system is related to the generation sources at specific buses. In the original system, certain buses had a mix of conventional and renewable generation capacities. In the updated model, these buses are now considered to have generation capacities sourced entirely from wind farms. This shift reflects the growing importance of renewable energy in modern power systems as explained in section 1. The total generation capacity at these buses remains unchanged; however, it is now assumed to be derived solely from renewable sources. Additionally, the minimum generation capacity at these buses has been adjusted to reflect the lowest conventional generation capacity that was present in the previous version. This ensures that the minimum generation level remains realistic while accounting for the variability inherent in renewable energy sources.

Additionally, the transmission network has been simplified in the revised model. In the modified ver-

sion, only 34 lines are considered. The extra lines, which created double connections between certain buses, have been removed to reduce complexity and focus the analysis on single connections between buses. This adjustment makes the model more efficient and easier to manage while still providing a realistic simulation of the power grid.

The 34 transmission lines in the revised system exhibit distinct characteristics that affect their operational rules within the network. Specifically, 17 of these lines have been designed with the capability for line switching, allowing them to be turned on or off based on operational requirements. This feature provides greater flexibility in managing the power flow and mitigating congestion within the network. In addition, 1 line is equipped with a phase angle regulator, which can be adjusted within a range of **-0.5 to 0.5**. This regulator allows for fine-tuning of the power flow across the network, helping to optimize the overall system performance. The remaining 16 lines follow the standard rules of the DC Optimal Power Flow (DCOPF) model, exhibiting no special characteristics.

It is important to note that the lines with line switching capabilities and those with PAR are mutually exclusive; a line cannot possess both attributes simultaneously. This separation of functionalities ensures that each line is utilized in the most efficient manner possible, depending on its specific role within the network.

To account for the increased integration of renewable energy and to avoid potential power imbalances, the revised model also introduces additional loads at several buses. Specifically, an extra 100 MW load is added to each of the buses within the range of bus numbers 3 to 10. This adjustment is intended to simulate the challenges posed by the high penetration of renewable energy sources in power systems. By introducing these additional loads, the model better reflects the real-world conditions under which modern power systems operate, where fluctuations in renewable generation can lead to significant changes in load demand.

These modifications collectively create a more realistic and contemporary version of the IEEE RTS 24-Bus System. The revised model emphasizes the role of renewable energy, simplifies the generation and transmission configurations, and introduces additional loads to account for the dynamic nature of modern power systems. By implementing these changes, the modified system offers a valuable tool for studying the impacts of high renewable energy integration on power system stability, congestion management, and overall operational efficiency. This updated version is more aligned with current trends in power systems, providing a robust framework for analyzing future energy scenarios and developing strategies for efficient power system management.

4.3 Identification of Congestion using Economic Dispatch and PTDF Method

The management of electricity generation and distribution involves several sophisticated methods to ensure efficiency and reliability. Among these methods, the Economic Dispatch Problem and the Power Transfer Distribution Factors (PTDF) method are prominent. This discussion elaborates on how these methods are implemented and the mathematical foundation supporting their application.

4.3.1 Economic Dispatch Problem

The Economic Dispatch Problem focuses on minimizing the cost of electricity generation while satisfying the demand for power. The objective function for this problem is represented as the sum of the products of the cost coefficients C_g^G and the power generated P_g^G for each generator g in the set of all generators G .

$$\text{Minimize } \sum_{g \in G} C_g^G p_g^G \quad (4.1)$$

This objective function seeks to find the least-cost combination of power outputs from all available generators that can meet the total demand.

The constraints associated with this problem ensure that the power system operates within its physical and operational limits. The first constraint is the power balance constraint, which ensures that the total power generated equals the total power demanded.

Power balance equation:

$$\sum_{g \in G} p_g^G = \sum_{b \in B} P_b^D \quad (4.2)$$

The second constraint is the generation capacity limit, which ensures that the power output of each generator does not exceed its maximum or fall below its minimum capacity.

Generation limit equation:

$$\sum_{g \in G} p_g^G = \sum_{b \in B} P_b^D \quad (4.3)$$

4.3.2 Power Transfer Distribution Factor Method

Following the determination of optimal power generation settings through the Economic Dispatch, the PTDF method is employed to analyze power flow in the transmission network and identify potential congestion. The PTDF method uses linearized power flow equations around a specified operating point, typically considering a particular bus as a reference. In IEEE RTS-24 system, Bus 13 is considered as reference bus [19].

To apply the PTDF method, the bus-line adjacency matrix $\hat{A}$ and the diagonal matrix of line reactances X are used [8]. The bus-line adjacency matrix is adjusted to exclude the reference bus, and the line flow vector p^L and the vector of bus angles δ are calculated by solving the following set of equations derived from Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) [8].

KVL equation:

$$Xp^L + \hat{A}^T \delta = 0 \quad (4.4)$$

KCL equation:

$$\hat{A}p^L = p^B \quad (4.5)$$

Combining these equations, the relationship is established in matrix form:

$$\begin{bmatrix} X & \hat{A}^T \\ \hat{A} & 0 \end{bmatrix} \begin{bmatrix} p^L \\ \delta \end{bmatrix} = \begin{bmatrix} 0 \\ p^B \end{bmatrix} \quad (4.6)$$

$$\begin{bmatrix} X & \hat{A}^T \\ \hat{A} & 0 \end{bmatrix} \begin{bmatrix} p^L \\ \delta \end{bmatrix} = \begin{bmatrix} 0 \\ p^B \end{bmatrix} \iff \begin{bmatrix} I & -X^{-1}\hat{A}^T \\ 0 & -\hat{A}X^{-1}\hat{A}^T \end{bmatrix} \begin{bmatrix} p^L \\ \delta \end{bmatrix} = \begin{bmatrix} 0 \\ p^B \end{bmatrix} \quad (4.7)$$

Given that all line reactances are non-zero, the matrix inversion is feasible, and the solutions for δ and p^L can be directly computed [8].

$$\delta = -(AX^{-1}A^T)^{-1}p^B = F^A p^B \quad (4.8)$$

$$p^L = X^{-1}A^T(AX^{-1}A^T)^{-1}p^B = F^P p^B \quad (4.9)$$

These PTDFs provide a direct measure of how power injections at various buses influence the flows on transmission lines, which is crucial for planning and operation in the context of system security and congestion management [8]. In this manner, Economic Dispatch and PTDF are implemented to effectively identify potential congestion in transmission lines.

4.4 Implementation of Congestion Management Actions

This section discusses the implementation of the congestion management strategies employed in the model. The strategies include load shedding, line switching, and the use of phase angle regulator.

Each technique plays a vital role in managing congestion within the power grid, ensuring that the system operates efficiently and reliably, even under conditions of high renewable energy penetration.

4.4.1 Load Shedding

Load shedding is the first strategy considered in the model. It is used when the demand for electricity is higher than what the system can supply. When this happens, some of the load at certain buses is reduced. This means that the electricity supply to certain areas is cut back to prevent the system from overloading. In this model, load shedding is applied to each bus with a load whenever it is necessary to balance the system.

Load shedding directly affects the consumers because it means they get less electricity than they need. Hence, it is not a desirable solution. Because of this, the cost of load shedding is set very high in the model, at **£45 per bus**. This high cost is justified by the significant inconvenience and potential economic losses that consumers may experience when their power supply is interrupted. It emphasizes the need to use load shedding only when absolutely necessary.

4.4.2 Line Switching

Line switching is another congestion management technique used in the model. This method involves turning certain transmission lines on or off to control the flow of electricity. Line switching is particularly useful for managing congestion in lines that are either already congested or are close to becoming congested. In this model, the lines that are congested are identified using the PTDF method as mentioned in previous section 4.3.2. Additionally, some other lines are also considered for line switching.

The cost of line switching is set relatively low, at **£5 per line** with switching functionality, because it is a straightforward and effective method to control power flow without requiring complex adjustments or significant interventions. The minimal operational effort required for line switching makes it a practical and cost-effective tool in the congestion management strategy.

4.4.3 Phase Angle Regulator

Phase angle regulators are used to control the flow of electricity between two parts of the power system. In this model, a PAR is installed on the line **between bus 3 and bus 24**. This line is a key point in the system because it divides the grid into two parts. One side, near bus 24, has more generation units, while the other side, near bus 3, has fewer generation units. By adjusting the phase angle with a PAR, the flow of electricity between these two parts of the system can be managed more effectively.

The cost of installing and using a PAR is set at **£35 per line** with PAR functionality. This cost is higher than that of line switching due to the sophisticated technology and specialized equipment required for its operation. However, it is carefully considered to reflect the long-term benefits and stability it provides in managing power flows. PAR are used when other methods like line switching are not enough to manage the congestion. Although the implementation of PAR involves a significant investment, they offer a strategic advantage in optimizing the power flow and maintaining grid reliability.

4.5 Data Compilation for Modified IEEE RTS 24-Bus System

This subsection presents the data that forms the foundation of the modified IEEE RTS 24-Bus System used in the study. The data provided here has been carefully adjusted to fit the modifications discussed in earlier subsection. This is done so that the system resembles a modern power grid with high renewable energy sources. The data includes details about transmission lines, bus loads and generators in the power system. Data presented herein are based on the original information from [19] and [28] papers. However, it has been altered to align with the modifications made to the system.

4.5.1 Transmission Line Data:

The system includes 34 transmission lines. The table shows detailed attributes of each line that are essential for accurately modeling power flows. The interconnection data for each transmission line includes the bus from which the line starts and the bus to which the line is connected. Each transmission line has a specified reactance and a maximum power flow capacity.

Transmission Line	From Bus	To Bus	Maximum Power Flow Capacity [MW]	Reactance [p.u.]	Operational Functionality
Line 1	1	2	175	0.0139	Switching
Line 2	1	3	175	0.2112	Switching
Line 3	1	5	350	0.0845	Switching
Line 4	2	4	175	0.1267	Ordinary
Line 5	2	6	175	0.192	Switching
Line 6	3	9	175	0.119	Ordinary
Line 7	3	24	400	0.0839	PAR
Line 8	4	9	175	0.1037	Ordinary
Line 9	5	10	350	0.0883	Ordinary
Line 10	6	10	175	0.0605	Switching
Line 11	7	8	350	0.0614	Ordinary
Line 12	8	9	175	0.1651	Ordinary
Line 13	8	10	175	0.1651	Switching
Line 14	9	11	400	0.0839	Ordinary
Line 15	9	12	400	0.0839	Ordinary
Line 16	10	11	400	0.0839	Ordinary
Line 17	10	12	400	0.0839	Ordinary
Line 18	11	13	500	0.0476	Switching
Line 19	11	14	500	0.0418	Ordinary
Line 20	12	13	500	0.0476	Ordinary
Line 21	12	23	500	0.0966	Switching
Line 22	13	23	250	0.0865	Switching
Line 23	14	16	250	0.0389	Switching
Line 24	15	16	500	0.0173	Ordinary
Line 25	15	21	400	0.049	Switching
Line 26	15	24	500	0.0519	Switching
Line 27	16	17	500	0.0259	Switching
Line 28	16	19	500	0.0231	Switching
Line 29	17	18	500	0.0144	Ordinary
Line 30	17	22	500	0.1053	Switching
Line 31	18	21	1000	0.0259	Switching
Line 32	19	20	1000	0.0396	Ordinary
Line 33	20	23	1000	0.0216	Switching
Line 34	21	22	500	0.0678	Ordinary

Table 1: Transmission line data

In this congestion management model, a transmission line can have only one of three operational functionalities: it can be either ordinary, equipped with line switching capability, or equipped with a PAR. Table 1 contains the connection details of the transmission lines, along with their maximum power flow capacity, reactance, and operational functionalities.

4.5.2 Demand Data

Table 2 contain bus load data, which is a representation of the total demand at each bus after making necessary changes. This data captures the changes made by addition of 100 MW of load connected to buses 3 through 10. This change accounts for increased demand after the addition of renewable energy sources to the system. The table will contain the modified load data that shows how power is consumed across a grid.

Load	Location	Power Demand [MW]
P_1^D	Bus 1	108
P_2^D	Bus 2	97
P_3^D	Bus 3	280
P_4^D	Bus 4	174
P_5^D	Bus 5	171
P_6^D	Bus 6	236
P_7^D	Bus 7	225
P_8^D	Bus 8	271
P_9^D	Bus 9	275
P_{10}^D	Bus 10	295
P_{13}^D	Bus 13	265
P_{14}^D	Bus 14	194
P_{15}^D	Bus 15	317
P_{16}^D	Bus 16	100
P_{18}^D	Bus 18	333
P_{19}^D	Bus 19	181
P_{20}^D	Bus 20	128
Total Demand		3650

Table 2: Demand data at time of system peak

4.5.3 Generator Data

Generators are the sources of power in the system. Each generator is connected to a specific bus and has its own unique characteristics. These characteristics include the type of generator, the costs of generation, the maximum generation capacity, and the minimum generation capacity. In the modified model, the generators at some buses have been combined into a single generator. At certain buses, only wind farms are now used as the power source, as explained in subsection 4.2. This change simplifies the model and focuses on renewable energy.

Generator	Location	Generator Type	Maximum Generation Capacity [MW]	Minimum Generation Capacity [MW]	Cost [£/MWh]
G1	Bus 1	OCGT	152	30.4	25
G2	Bus 2	OCGT	152	30.4	25
G3	Bus 3	Wind	200	0	10
G5	Bus 5	Wind	200	0	10
G7	Bus 7	Wind	550	75*	10
G13	Bus 13	Coal	591	206.85	30
G15	Bus 15	CCGT	215	12	20
G16	Bus 16	Wind	355	54.25*	10
G18	Bus 18	CCGT	400	100	20
G21	Bus 21	Wind	600	100*	10
G22	Bus 22	Coal	300	300	30
G23	Bus 23	Wind	860	108.5*	10

* Non-zero Minimum generation from wind sources account for the conventional generation capacity in previous version of the IEEE 24 bus system.

Table 3: Data for generator units

The costs for generating power have been adjusted to make the model easier to implement. These costs are assumed and hypothetical, used mainly to show how the congestion management technique can maximize power generation from wind and other renewable sources. In this model, the cost of generating power from wind is set as the lowest because the aim is to integrate renewable energy into the power grid while managing congestion and keeping the total costs down. The other generation costs are relative and have been chosen to reflect different types of power generation.

Table 3 provides details about the generator units in the system. It shows the bus location of each generator, the type of generator, the generation costs and the maximum and minimum generation capacities of the generators [19][28].

5 Results and Analysis

Results from the application of the economic dispatch problem and PTDF analysis on the modified IEEE RTS 24-Bus system reveal key insights into potential congestion areas. This section delves into the quantitative outcomes, followed by an in-depth analysis of the proposed congestion management strategies. The reference bus for this study is Bus 13 [19]. The effectiveness of these strategies in optimizing operating costs and enhancing renewable energy penetration is critically assessed, providing a holistic view of the system's performance.

5.1 Economic Dispatch Problem

The economic dispatch analysis was conducted to determine the optimal power generation allocation across the various generators in the modified IEEE RTS 24-Bus system. The results, presented in 4, include the power output of each generator, the operating cost associated with each generator, and the total operating cost of the system.

Generator	Power Generation [MW]	Operating Cost [£]
G1	30.4	760
G2	30.4	760
G3	200	2000
G5	200	2000
G7	550	5500
G13	206.85	6205.5
G15	12	240
G16	355	3550
G18	305.35	6107
G21	600	6000
G22	300	9000
G23	860	8600
Total Operating Cost		50722.5

Table 4: Power generation and operating costs for economic dispatch

The power generation by all six renewable energy sources (G3, G5, G7, G16, G21 and G23) contribute the maximum generation due to their significantly lower generation costs (£/MWh) compared to other sources. This outcome aligns with the objective of maximizing renewable energy integration while minimizing the overall system operating costs. The economic dispatch effectively prioritizes the use of renewables, resulting in a cost-efficient allocation of generation resources.

5.2 PTDF Method

Table 5 showcases the results from implementing the Power Transfer Distribution Factor (PTDF) method on the modified version of the IEEE RTS 24-Bus system. This method helps identify how electricity flows through the transmission lines and provides insight into potential areas where the power system might become congested. This table is laid out to provide detailed insights into the power flow across various transmission lines, each uniquely identified for clarity and comprehensive analysis.

The table is neatly organized into several columns. Each row corresponds to one of the 34 transmission lines in the system. The first column lists the transmission lines by number for easy reference. The second and third columns show the starting and ending bus numbers for each line, indicating the path that electricity takes from one part of the network to another. The fourth column, displays the amount of power, in megawatts, flowing through each line. This column is essential as it shows whether the power flow on a particular line is within the system's capacity or if it exceeds it. These instances of overload are critical as they highlight areas within the network where congestion is likely, affecting

overall system efficiency and reliability.

Transmission Line	From Bus	To Bus	Power Flow by PTDF method [MW]	Location	Voltage Angle
Line 1	1	2	93.39	Bus 1	-35.96
Line 2	1	3	-106.21	Bus 2	-37.26
Line 3	1	5	-64.78	Bus 3	-13.53
Line 4	2	4	9.58	Bus 4	-38.48
Line 5	2	6	17.21	Bus 5	-30.49
Line 6	3	9	66.34	Bus 6	-40.57
Line 7	3	24	-252.55	Bus 7	0.04
Line 8	4	9	-164.42	Bus 8	-19.92
Line 9	5	10	-35.78	Bus 9	-21.43
Line 10	6	10	-218.79	Bus 10	-27.33
Line 11	7	8	325.00	Bus 11	-6.93
Line 12	8	9	9.12	Bus 12	-5.39
Line 13	8	10	44.88	Bus 13	0
Line 14	9	11	-172.81	Bus 14	4.38
Line 15	9	12	-191.15	Bus 15	20.76
Line 16	10	11	-243.17	Bus 16	22.45
Line 17	10	12	-261.52	Bus 17	33.12
Line 18	11	13	-145.52	Bus 18	36.51
Line 19	11	14	-270.46	Bus 19	20.01
Line 20	12	13	-113.18	Bus 20	23.01
Line 21	12	23	-339.49	Bus 21	43.32
Line 22	13	23	-316.85	Bus 22	51.70
Line 23	14	16	-464.46	Bus 23	27.41
Line 24	15	16	-97.17	Bus 24	7.66
Line 25	15	21	-460.37		
Line 26	15	24	252.55		
Line 27	16	17	-411.98		
Line 28	16	19	105.34		
Line 29	17	18	-235.51		
Line 30	17	22	-176.47		
Line 31	18	21	-263.16		
Line 32	19	20	-75.66		
Line 33	20	23	-203.66		
Line 34	21	22	-123.53		

Table 5: Power flows and voltage angles by PTDF method

Accompanying the power flow data, the *Location* column specifies the bus associated with each entry in the *Voltage Angle* column. Voltage angles are key to understanding the electrical behavior within the network, such as phase differences that influence the direction and magnitude of power flows between buses. Large differences in voltage angles across the network indicate potential stress or instability areas, necessitating careful management to ensure system stability.

Several lines exhibit power flows that exceed their maximum flow capacity, signaling potential congestion points. For example, Line 23 shows a significant overload with a power flow of -464.46 MW against a capacity of 250 MW. Similarly, Line 10, Line 22 and Line 25 each show power flows of -218.79 MW, -316.85 MW and -460.37 MW that violate the respective flow capacities, indicating severe stress and potential risk to system stability. This makes these lines prime candidates for line switching. In contrast, some lines such as Line 4 with a power flow of 17.21 MW and Line 12 with 9.12 MW are far below their capacity limits, suggesting they are underutilized. This can point to inefficiencies in power distribution and opportunities for optimizing the flow to better balance the network load. The line flow limit violation for lines **22, 23, 25 and 27** is observed from table 5. This suggests that these

lines are congested and susceptible to failure.

The voltage angles give insights into the phase difference across the network, which affects the direction and rate of power flow between buses. Large variations can indicate potential issues. For instance, Bus 1 shows a voltage angle of -35.96 degrees, while nearby Bus 2 is at -37.26 degrees, demonstrating a small phase difference and stable conditions between these two points. Whereas, Bus 9 and Bus 11 show -21.43 and -6.93 voltage angles respectively, which indicates instability.

5.3 MILP for Modified IEEE RTS 24-Bus System

This section presents the findings from the implementation of the congestion management model described in section 3. The Mixed-Integer Linear Programming (MILP) model is used to explore different scenarios for managing congestion in the modified RTS power grid. The results from this model include data on total operational costs, power generation by all generators, power flow through all 34 lines, and the voltage phase angles for all 24 buses. These results are presented for eight different cases, each employing various combinations of congestion management techniques.

5.3.1 Total Operating Costs

Table 6 outlines the total operational costs associated with different combinations of congestion management techniques applied in a DC power system. This table is organized into three columns and includes data for eight distinct cases, each representing a different set of strategies utilized to manage congestion. The scenarios range from using no management techniques to deploying all three available methods: Load Shedding, Line Switching, and Phase Angle Regulator.

Case No.	Congestion Management Action Status			Total Cost [£]
	Load Shedding ($w_1 = 1$)	Line Switching ($w_2 = 1$)	PAR ($w_3 = 1$)	
1	-	-	-	59908.140
2	Active	-	-	57367.114
3	-	Active	-	56349.856
4	-	-	Active	59365.188
5	Active	Active	-	55295.940
6	Active	-	Active	57342.379
7	-	Active	Active	56334.826
8	Active	Active	Active	55278.412

Table 6: Total operating cost for combinations of congestion management techniques

The first column in Table 6 sequentially numbers the cases, providing an easy reference point for each unique combination of management strategies examined in the study. The second column is subdivided into three sub-columns, each representing one of the three congestion management actions studied: Load Shedding ($w_1 = 1$), Line Switching ($w_2 = 1$), and Phase Angle Regulator ($w_3 = 1$). Each sub-column indicates whether the specific action is active in the corresponding case, denoted by the word **Active** or not utilized indicated by a dash (-). The last column lists the total operational cost associated with each case, measured in pounds sterling (£). The costs reflect the combined expenses of power generation and implementing congestion management actions.

The **baseline scenario**, labeled as **Case 1**, involves no congestion management actions. That means it utilizes the standard DCOPF setup, resulting in an operating cost of **£59,908.14**. The baseline scenario serves as a reference point to gauge the effectiveness of each congestion management strategy in terms of cost reduction.

The analysis of the different congestion management techniques reveals varying degrees of improvement in operating costs. In Case 2, where only load shedding was applied, there was a modest improvement

with a 4.24% reduction in costs. This indicates that while effective, load shedding alone may not significantly optimize system operations. Case 3, which utilizes only line switching, demonstrated a more substantial improvement with a 5.94% reduction. This highlights the effectiveness of this technique in reducing operational expenses. In contrast, Case 4, which applied only PAR, resulted in a minimal cost reduction of 0.91%. This suggests its impact on cost reduction is limited when used independently.

When techniques were combined, further improvements were observed. In Case 5, the combination of load shedding and line switching led to a significant 7.70% decrease in operating costs. This demonstrates the synergistic effect of these two techniques. Case 6, which paired load shedding with PAR, achieved a 4.28% reduction. This shows an improvement over using PAR alone but less effective than the combination in Case 5. Similarly, in Case 7, the combination of line switching and PAR yielded a 5.96% reduction, reinforcing the strong contribution of line switching to cost optimization even when used alongside PAR.

Finally, in **Case 8**, where all three techniques were employed together, the analysis revealed the greatest **overall improvement** with a **7.73%** reduction in operating costs. This indicates that using all three techniques together provides the greatest overall cost savings. This comprehensive approach leverages the unique capabilities of each technique to reduce costs effectively.

5.3.2 Power Generation Output

Table 7 meticulously details the power generation outputs from various generators under eight different congestion management scenarios. The table highlights the shift in generation pattern that corresponds to changes in management strategies. The generators are categorized into different types, including OCGT (G1, G2), wind (G3, G5, G7, G16, G21, G23), coal (G13, G22), and CCGT (G15, G18), as previously outlined in section 4.5 of the data compilation. This differentiation is crucial as it sheds light on how renewable sources, respond to congestion management techniques compared to conventional sources.

Generator Type	Power generation for the Case mentioned in columns [MW]							
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
G1	152	152	138.314	152	152	152	137.227	152
G2	152	152	152	152	152	152	152	152
G3	200	200	200	200	200	200	200	200
G5	200	200	200	200	200	200	200	200
G7	428.931	550	550	455.636	550	550	550	550
G13	581.995	324.497	418.007	552.654	350.177	323.750	417.196	349.848
G15	20.824	12	12	23.460	12	12	12	12
G16	54.250	122.441	198.275	54.250	355	123.968	200.895	355
G18	100	100	100	100	100	100	100	100
G21	600	600	526.894	600	575.761	600	526.351	575.376
G22	300	300	300	300	300	300	300	300
G23	860	860	854.510	860	700.137	860	854.330	701.663

Table 7: Power generation at the time of system peak

The next columns in Table 7 correspond to each of the eight cases analyzed, mirroring the structure seen in Table 6. Each column shows the power output of every generator for the specific case, allowing for a direct comparison of how different management strategies affect the generation output of each type of generator. The power output from various generators exhibits notable fluctuations depending on the congestion management strategy employed. Generators such as G7 and G13 display significant variability in output, suggesting their sensitive response to different grid conditions induced by varying management actions.

Wind generators, which include G3, G5, G7, G16, G21, and G23, generally maintain stable or increased output across most cases, particularly when congestion management strategies are applied.

This stability is indicative of the effectiveness of varied strategies in supporting integration of sustainable options. Notably, G7 and G21 show increased generation in cases where load shedding and line switching are active, underscoring the potential of these techniques to enhance wind energy contribution. OCGT and coal generators, on the other hand, exhibit less variability compared to wind generators. For instance, G1 and G2 (OCGT) consistently produce the same level of output across all cases, showing minimal sensitivity to congestion management strategies.

Case 1, serving as the baseline, shows the highest total generation cost at £59,908.14. This indicates a less efficient operation of the system, highlighting the potential benefits of implementing management strategies to optimize power distribution and cost.

Moving to Case 2, where load shedding is the sole technique used, there is a significant reduction in total generation cost to £53,899.32. This substantial decrease underscores the effectiveness of load shedding in curtailing unnecessary power generation, particularly from costlier conventional sources. Hence enhancing overall system efficiency and reducing expenses.

The scenarios involving line switching (Cases 3, 5, 7 and 8) demonstrate further reductions in total generation costs. Case 8 achieving the lowest cost of £55,155.83 when all techniques are employed simultaneously. This trend suggests that line switching plays a crucial role in rerouting power to alleviate congested pathways. This not only supports the integration of renewable generation but also significantly lowers operational costs by improving the overall flow within the power grid.

The influence of phase angle regulator, as observed in Cases 4, generally results in higher costs when used alone. However, when PAR is combined with other management techniques (Cases 6, 7 and 8), the overall costs are optimized. This outcome reveals the nuanced role of PAR in managing grid dynamics. While PAR may not always be cost-effective on its own; its integration with other techniques such as load shedding and line switching can lead to cost efficiency in the grid's operation.

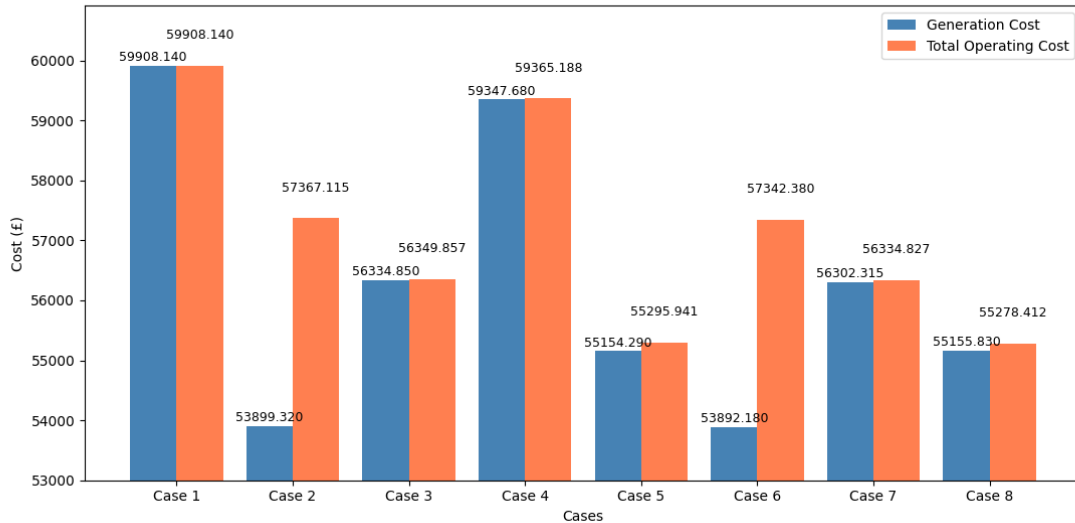


Figure 2: Total Generation Cost vs Total Operating Cost across different cases

The fig. 2 represents the comparison between the total generation cost and total operating cost of the system in each case. By observation, the minimum generation cost corresponds to Case 2. However, Case 2 is not the best solution among all eight cases. This suggests load shedding causes higher penalties.

5.3.3 Load Shedding Results

The model considers k_b as the variable for proportion of load supplied at each bus. In case of load shedding, few buses experience reduction in demand supply.

In Case 2, load shedding occurs at bus 14, where only 60.3% of the bus's load of 194 MW is met. Consequently, the incurred penalty for unmet load is calculated as follows:

$$\begin{aligned}\text{Penalty} &= \text{Cost of load shedding} \times \text{Unmet load} \\ &= 45 \times (1 - 0.603) \times 194 \\ &= 3465.81\end{aligned}$$

In Case 5, which combines load shedding with line switching, bus 10 undergoes load shedding, receiving 99% of its 295 MW load. The resulting penalty for this slight shortfall amounts to £132.75. During Case 6, where load shedding is coupled with the use of phase angle regulators, bus 14 is again subjected to load shedding, this time receiving 60.7% of its load, leading to a penalty of £3430.89. In Case 8, which integrates all three congestion management techniques, bus 10 achieves better load fulfillment at 99.3%, reducing the penalty to a mere £92.93.

The results indicate that integrating multiple congestion management strategies significantly reduces the penalties associated with unmet load demands. Specifically, the application of all three techniques in Case 8 demonstrates the most effective reduction in penalties, suggesting a synergistic effect of the combined strategies in minimizing the economic consequences of load shedding.

5.3.4 Transmission Line Power Flow Results

In the investigation of congestion management techniques within the power grid, the strategic deactivation of specific transmission lines plays a pivotal role in optimizing system performance and reducing operating costs. The candidates for line switching, as determined by the PTDF method, include lines 10, 22, 23 and 25. Along with them few other lines such as lines 1, 2, 3, 5, 13, 18, 21, 26, 27, 28, 30, 31, and 33 are also selected as candidates for line switching. Among all these candidate lines, the model chooses lines to be switched off based on their potential to alleviate congestion effectively.

The results from the model's outputs indicate which lines are deactivated in various scenarios, highlighting the adaptability and strategic focus of congestion management. The model's results for various cases illustrate the selective deactivation of these lines:

- Case 3: lines 13, 28, and 30 are switched off.
- Case 5: lines 18 and 23 are switched off.
- Case 7: lines 13, 28, and 30 are switched off.
- Case 8: lines 18 and 23 are switched off.

Analyzing the impact of these deactivations on operating costs reveals significant insights. For instance, the deactivation of lines 18 and 23 in both Cases 5 and 8 correlates with substantial reductions in operating costs. Specifically, Case 8, where these lines are turned off alongside other strategic adjustments, results in the lowest operating costs observed among all scenarios. The strategic management of these lines not only helps in balancing load across the network but also underscores their critical role in maintaining economic efficiency within the system's operation.

Table 8 provides a clear depiction of how congestion management techniques influence the power flow in transmission lines. The analysis particularly focuses on lines 10 and 23, which were identified as critical for potential congestion and failure by PTDF method. From the baseline scenario (Case 1), where Line 10 starts with a power flow of -175 MW and Line 23 begins at -250 MW, we

Line	Power flow for the Case mentioned in columns [MW]							
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
1	74.389	75.201	64.858	74.389	104.844	75.474	64.672	105.029
2	-28.061	-29.75	-31.449	-28.061	-91.12	-30.317	-32.336	-91.574
3	-2.328	-1.451	-3.094	-2.328	30.276	-1.157	-3.109	30.545
4	68.389	68.646	58.858	68.389	78.208	68.732	58.672	78.219
5	61	61.555	61	61	81.636	61.741	61	81.81
6	39.272	42.862	25.72	39.272	173.757	44.068	26.911	174.607
7	-147.333	-152.612	-137.169	-147.333	-344.877	-154.385	-139.246	-346.181
8	-105.611	-105.354	-115.142	-105.611	-95.792	-105.268	-115.328	-95.781
9	26.672	27.549	25.906	26.672	59.276	27.843	25.891	59.545
10	-175	-174.445	-175	-175	-154.364	-174.259	-175	-154.19
11	203.931	325	325	230.636	325	325	325	325
12	-43.866	16.423	54	-30.514	7.739	16.341	54	7.615
13	-23.203	37.577	0	-9.85	46.261	37.659	0	46.385
14	-162.885	-140.632	-123.384	-156.457	134.901	-139.956	-122.905	135.147
15	-222.321	-180.436	-187.039	-215.396	-324.197	-179.902	-186.513	-323.706
16	-203.547	-182.258	-190.219	-197.119	59.099	-181.905	-190.251	58.853
17	-262.983	-222.062	-253.875	-256.059	-400	-221.851	-253.858	-400
18	-310.432	-189.828	-257.603	-297.576	0	-189.579	-257.156	0
19	-56	-133.062	-56	-56	194	-132.283	-56	194
20	-205.67	-119.669	-145.404	-193.69	-335.177	-119.17	-145.04	-334.848
21	-279.634	-282.829	-295.51	-277.765	-389.021	-282.583	-295.33	-388.859
22	-199.107	-250	-250	-203.612	-250	-250	-250	-250
23	-250	-250	-250	-250	0	-250	-250	0
24	-60.725	-72.874	-42.169	-58.454	-249.877	-74.401	-44.246	-251.181
25	-382.785	-384.738	-400	-382.42	-400	-384.984	-400	-400
26	147.333	152.612	137.169	147.333	344.877	154.385	139.246	346.181
27	-284.215	-282.262	-193.894	-284.58	-242.761	-282.016	-193.351	-242.376
28	-72.26	-18.171	0	-69.623	247.884	-18.417	0	246.195
29	-106.949	-105.365	-193.894	-107.245	-73.323	-105.165	-193.351	-73.011
30	-177.266	-176.897	0	-177.335	-169.437	-176.851	0	-169.365
31	-339.949	-338.365	-426.894	-340.245	-306.323	-338.165	-426.351	-306.011
32	-253.26	-199.171	-181	-250.623	66.884	-199.417	-181	65.195
33	-381.26	-327.171	-309	-378.623	-61.116	-327.417	-309	-62.805
34	-122.734	-123.103	-300	-122.665	-130.563	-123.149	-300	-130.635

Table 8: Power flow in transmission lines at the time of system peak

observe that both lines operate near their capacity limits, highlighting potential congestion challenges.

In Case 2, implementing load shedding slightly alters the power flow in Line 10 to -174.445 MW while Line 23 remains unchanged at -250 MW, suggesting that load shedding alone does not effectively address the congestion in these lines. However, in Case 3 with line switching employed, Line 10 remains at -175 MW, but Line 23’s power flow is reduced to 0 MW, indicating that the model chooses to switch off this line to effectively manage congestion.

When PAR is introduced in Case 4, there’s no change in the power flows of Line 10 and Line 23 from the baseline scenario, indicating a limited impact of PAR on its own. The combination of load shedding and line switching in Case 5 brings a more significant reduction, with Line 10’s power flow dropping to -154.364 MW and Line 23 being switched off again, showcasing effective congestion management.

Continuing this trend in Case 6, which combines load shedding with PAR, the power flow in Line 10 adjusts slightly to -174.259 MW, and Line 23 is kept switched off, maintaining effective congestion relief. In Case 7, the combination of line switching and PAR results in similar management outcomes, with Line 23 remaining switched off.

The most comprehensive approach in Case 8, where all techniques are utilized, leads to a further reduction in Line 10 to -154.19 MW and keeps Line 23 switched off, illustrating the peak effectiveness of integrating all available management strategies. These cases demonstrate the critical role of targeted line management, particularly the strategic deactivation of lines like Line 23, in enhancing grid reliability and operational efficiency. This strategy not only secures the system against potential overloads but also ensures system stability.

5.3.5 Interpretation of Voltage Angles

Table 9 provides a detailed overview of the voltage phase angles at each bus during the system peak across the eight different scenarios. The first column identifies each of the 24 buses in the network, while the subsequent columns display the voltage phase angles in degrees for each case.

Bus	Voltage angle for the Case mentioned in columns							
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
1	-29.696	-22.017	-26.195	-29.465	-41.722	-21.925	-26.179	-41.66
2	-30.73	-23.063	-27.097	-30.499	-43.179	-22.974	-27.078	-43.12
3	-23.769	-15.734	-19.553	-23.539	-22.477	-15.522	-19.35	-22.319
4	-39.394	-31.76	-34.554	-39.164	-53.088	-31.683	-34.512	-53.03
5	-29.499	-21.895	-25.934	-29.269	-44.28	-21.827	-25.916	-44.241
6	-42.442	-34.881	-38.809	-42.211	-58.853	-34.828	-38.79	-58.827
7	-23.164	1.832	6.257	-22.164	-21.922	1.887	6.318	-21.886
8	-35.685	-18.123	-13.698	-35.014	-41.877	-18.068	-13.637	-41.841
9	-28.443	-20.835	-22.614	-28.212	-43.155	-20.766	-22.552	-43.098
10	-31.854	-24.327	-28.221	-31.624	-49.514	-24.286	-28.203	-49.499
11	-14.777	-9.036	-12.262	-14.654	-54.473	-9.024	-12.241	-54.437
12	-9.79	-5.696	-6.921	-9.676	-15.954	-5.673	-6.904	-15.939
13	0	0	0	0	0	0	0	0
14	-12.436	-3.474	-9.921	-12.313	-62.582	-3.495	-9.9	-62.546
15	-3.761	4.99	-0.926	-3.631	24.357	4.943	-0.94	24.192
16	-2.711	6.251	-0.196	-2.588	28.68	6.23	-0.175	28.537
17	4.65	13.562	4.826	4.775	34.967	13.535	4.833	34.815
18	6.19	15.079	7.618	6.316	36.023	15.049	7.617	35.866
19	-1.042	6.671	7.783	-0.931	22.954	6.656	7.783	22.85
20	8.988	14.558	14.951	9.077	20.305	14.553	14.951	20.268
21	14.995	23.843	18.674	15.122	43.957	23.808	18.66	43.792
22	23.317	32.189	39.014	23.442	52.809	32.157	39	52.649
23	17.223	21.625	21.625	17.301	21.625	21.625	21.625	21.625
24	-11.408	-2.93	-8.045	-11.278	6.458	-3.069	-8.167	6.225

Table 9: Voltage angles at the time of system peak

Analyzing the voltage angles provided in Table 9 across all eight cases offers significant insights into the stability and efficiency of the power grid during various congestion management strategies. Each case illustrates how voltage angles respond to the application of different congestion management techniques such as load shedding, line switching, and phase angle regulator. Bus 13 is set at 0 degrees, serving as the reference bus across all scenarios in model implementation [19]. This normalization helps in assessing the relative phase angles of other buses. Buses 1 through 12 and 14 through 24 show varying angles depending on the management strategies applied in each case, highlighting their responsiveness to system changes intended to alleviate congestion and optimize power flow.

The difference in phase angles between two distant buses can reflect the relative stress on the grid, even if those buses are not linked by a single transmission line.

The voltage angles in Case 1 represent the system under normal operating conditions without any congestion management interventions. Angles are spread within a range, with some buses like Bus

22 showing a relatively higher angle of 23.317 degrees, suggesting potential points of higher electrical pressure.

Case 2 shows a general reduction in the magnitude of voltage angles across most buses compared to the baseline, indicating a decrease in system stress due to reduced load. In Case 3, some buses like Bus 8 and Bus 9 exhibit significant shifts in their voltage angles, demonstrating the direct impact of deactivating certain lines. Subtle changes in voltage angles are noticeable in Case 4, reflecting the fine-tuning capability of PAR in adjusting the phase angles to optimize flows without drastic interventions.

Case 5 scenario shows a notable reduction in voltage angles, across all buses. The application of both load shedding and line switching strategically alleviates stress on the system, resulting in improved stability across these buses. Voltage angles in Case 6 are uniformly reduced across the network, similar to Case 2. This uniformity highlights the effective combination of load shedding with the fine-tuning capabilities of PAR, contributing to enhanced overall system stability. Similarly, Case 7 aligns closely with Case 3 but with additional fine-tuning provided by PAR, evident from the slight variations in voltage angles which help maintain a balanced network.

Lastly, Case 8 displays the most optimized set of voltage angles, with many buses showing reduced and more uniform angles across the network. This scenario indicates excellent system stability and efficient power distribution facilitated by the comprehensive application of all available congestion management techniques.

5.3.6 Optimized Network Topology of IEEE RTS 24-Bus System

The diagram illustrating the network topology for Case 8, shown in fig. 3, captures the optimized configuration resulting from the application of comprehensive congestion management techniques.

A clear indication of power generated by each generator on the power flow diagram shows how energy is distributed across the network. Deactivated transmission lines are removed in the model in fig. 3. The diagram depicts a streamlined and well managed network. The above representation is a clear description of the most effective operational state of the network achieved by the use of all available tools for congestion management.

Removal of transmission lines 18 and 23, simplifies the visual understanding of the network and highlights the strategic decisions made to optimize power flow through the system. This refined network topology presents a model of optimal performance and reliability in power system operation.

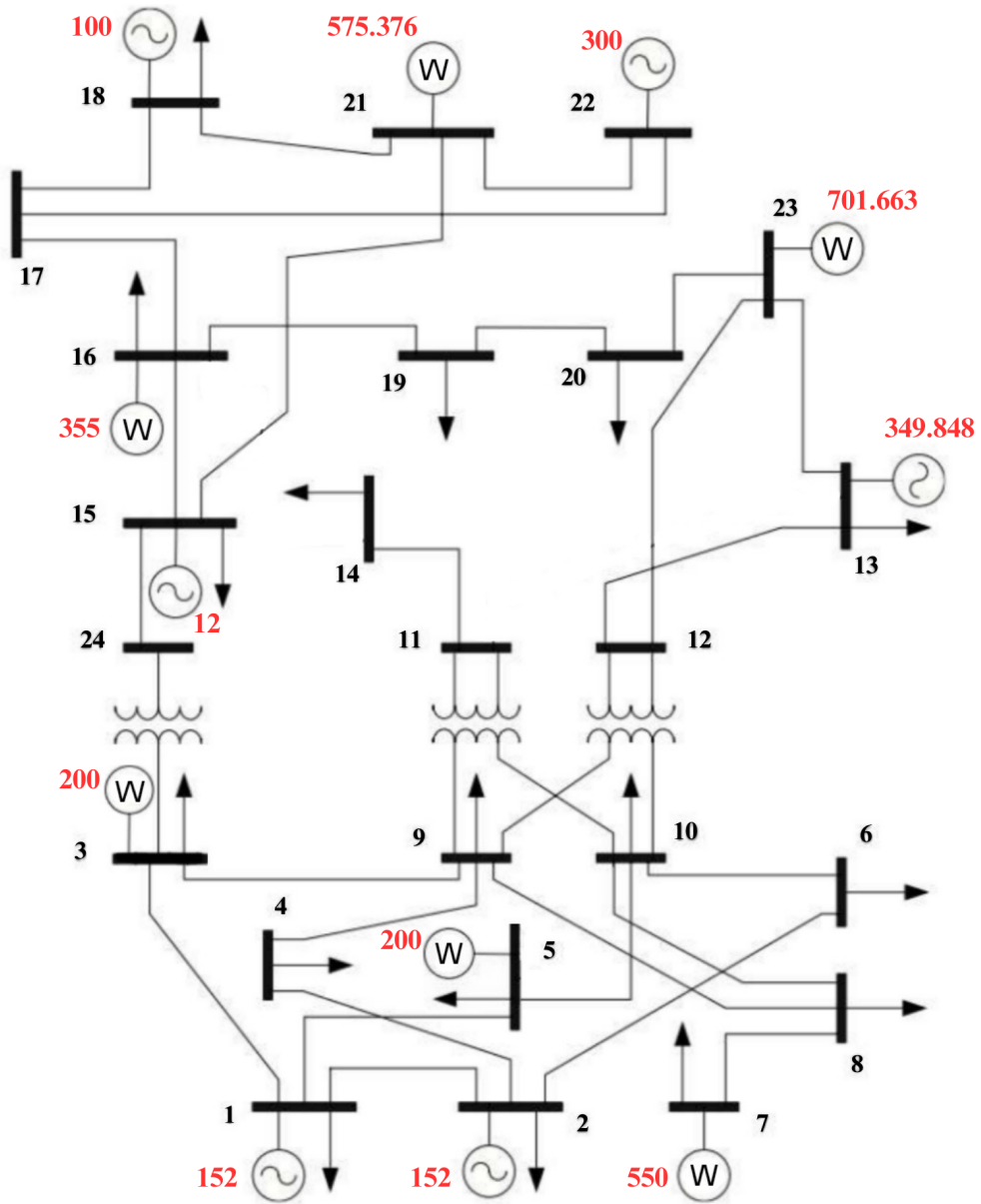


Figure 3: Optimized 24 bus system - Single area RTS-96

6 Conclusions and Future Works

6.1 Conclusion

The thesis presents a mathematical problem formulation incorporating load shedding, line switching, and phase angle regulator as techniques to handle congestion within the modified IEEE RTS 24-Bus System. It critically investigates the impact of such strategies on the power system using detailed numerical experiments. Data as presented in Tables 4 to 9 offers an in-depth analysis on power generation, operating costs, power flows, and voltage angles at different operational scenarios. Thus, providing ample evidence of the effectiveness of each approach to congestion management.

The detailed analysis presented in Tables 6 to 9 shows that each technique for congestion management has its benefits and limitations. Load shedding forms an important aspect in the relief from peak demand pressures. This effectively brings down generation costs as indicated in fig. 2. This approach effectively balances the system's load during peak demand periods, preventing undue strain on resources.

In contrast, line switching offers a powerful means of directly controlling electricity flows to prevent overloads, ensuring a more balanced distribution of power across the network. According to Table 8, through strategic deactivation of lines, the model prevents instability and possible failures.

Phase angle regulators have a subtle effect when used alone. However, they become important tools for fine-tuning system performance when combined with other techniques. As in Table 9, the PAR contributes to making the voltage angles remain controlled in a precise manner so that the power grid remains in balance and steady.

The role of Mixed Integer Linear Programming (MILP) has been instrumental in this study. By means of MILP, the congestion management techniques are assessed to successfully incorporate renewable energy sources into the power grid. MILP offers a sound framework for simulating several scenarios and analyzing potential outcomes. This modeling approach allows for the optimized combination of different congestion management strategies to improve system efficiency and reliability of the 24-bus system.

This paper tests the applicability of many methods, among which a strategy that combines the three techniques is found to be most effective in ensuring the lowest operating cost with optimal system performance as shown in Table 6. This all-inclusive approach not only ensures the operational efficiency of the power system but also enhances its ability to withstand and adapt to varying load conditions.

The study hence confirms that the multi-faceted congestion management approach using coordination between load shedding, line switching, and PAR provides the optimal solution for effectively managing power system congestion. This approach optimizes the power generation from renewable resources, thereby enhancing the overall sustainability of the grid.

6.2 Future Work

In this section, potential areas for future research are identified, building upon the work presented in this study to further enhance congestion management strategies in power systems.

1. **Multi-period Model:** Future research could explore extending the current model to a multi-period framework, allowing for the consideration of time-dependent variations in demand, generation, and system conditions.
2. **Extending the Problem Formulation to AC:** An important next step would be to adapt the current DC-based model to Alternating Current (AC) systems, which are more commonly used in large-scale power grids.

3. **Implementation on Real Power Grids:** Applying the developed model to real-world power grids would validate its practicality and effectiveness, moving beyond the standard test systems used in this study.
4. **Exploring Alternative Congestion Management Strategies:** Future work could also explore the inclusion of additional congestion management techniques, such as dynamic thermal line rating, bus splitting, and other advanced strategies, to further enhance the model's flexibility and applicability.

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Appendices

A Appendix

To see the code and additional results, please check the following GitHub Repository:
<https://github.com/rasikadalvi/Optimal-congestion-management-in-power-systems>